Dinghao Luo, PhD

PhD in Systems Neuroscience

Contact

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Technical Skills

Programming

Python (pandas, NumPy, matplotlib, seaborn)
SQL for data analysis
C++ (Arduino)
R, MATLAB

Machine Learning

PyTorch, scikit-learn
Image segmentation
Regression & classification

Data & Software

Git, GitHub
Automated testing, CI
Scientific data pipelines
GPU-accelerated computing (CuPy)

AI Tools

Ollama, Claude Code, Codex

Languages

Mandarin (native)
Cantonese (native)
English (fluent)
German (intermediate)

Awards & Honours

Jardine Scholarship
Full competitive funding at Cambridge University

IMPRS Fellow
Ranked amongst the top incoming candidates

Pingshan Book Award
Awarded for the Chinese edition of *Entangled Life*

Bundy Scholarship
Awarded for graduating with First Class at Cambridge

Systems neuroscientist experienced in turning multimodal biological data into reproducible analyses, statistical models, and usable Python tools.

Education

2021–2026 **PhD · Max Planck Florida Institute for Neuroscience**
2021–2023 **MS · Max Planck Florida Institute for Neuroscience**
2017–2020 **BA (Hons), First Class · University of Cambridge**

Work & Research Experience

MAX PLANCK FLORIDA INSTITUTE FOR NEUROSCIENCE, Jupiter, FL, US
PhD Researcher, Computational and Systems Neuroscience | 2021–2026

- Built statistical models and analytical tools for terabyte-scale neural and behavioural data; tested competing explanations against controlled experiments and produced a first-author manuscript under review at *Neuron*. [Preprint].
- Revised models and analyses when results contradicted the original framework.
- Diagnosed timing and data-integrity failures in synchronised acquisition systems spanning behaviour, imaging, and electrophysiology.

UNIVERSITY OF CAMBRIDGE, Cambridge, UK
Undergraduate Researcher, Neuroscience | 2019–2020

- Designed, performed, and analysed controlled biological experiments.
- Received First-Class marks for the resulting undergraduate thesis.

UNIVERSITY COLLEGE LONDON, London, UK
Research Intern, Biology | 2018

- Awarded a Genetics Society Studentship to study Alzheimer's Disease and Down syndrome; co-authored a peer-reviewed paper.

Selected Projects

FIBRESIGHT | *Image segmentation, computer vision & scientific software* [GitHub]

- Released a PyTorch image-segmentation workbench trained on 1,296 curated ROIs; achieved 0.93 recall and 0.83 F2 on nine held-out sessions, reducing manual curation from days to hours.
- Built a microscopy pipeline from raw images to neural activity; on a 40-case synthetic benchmark, FibreSight had a lower mean held-out 95th-percentile displacement error than Suite2p.

KAIROS | *Temporal modelling and research analytics* [GitHub]

- Built a Python pipeline with temporally lagged features and expanding-window validation; flexible time and publication-volume baselines explained nearly all apparent signal, so the analysis did not support additional predictive value.

Communication & Mentoring

- **Science writer and editor** at *Scientific American China* and Neu-Reality; interviewed prominent researchers for a feature on theories of consciousness.
- **Mentored** nine researchers and technicians in Python, statistics, data analysis and experimental workflows.
- **Translator** of *Entangled Life* (Sheldrake, 2020; Chinese ed. 2024; Pingshan Award, 2025) and *The Mind Within the Brain* (Redish, 2013; Chinese ed. forthcoming 2026).